MaleVis Related-Work Table and Research Gap Analysis

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Related Work: Malevis with CNN and Attention

The MaleVis dataset represents malware and benign software classes as three-channel RGB color images so that recurring byte-level visual patterns can be learned directly by image-classification models. The studies below were selected because they use convolutional neural networks, channel or spatial attention, or hybrid attention mechanisms tested on MaleVis or similar multi-class image configurations. This review focuses on the model designs, attention mechanics, evaluation protocols, reported accuracies, and limitations that are directly relevant to a Custom CNN + Attention framework, with a major focus on handling the redundant byte-image problem.

1. Al-Harbi & AlShorman (2026) introduced an A-HRAB framework incorporating an Adaptive Hybrid Residual Attention Block designed for visual malware family classification. The attention module dynamically balances channel and spatial attention maps using learnable weight parameters, allowing the network to emphasize descriptive binary textures in the code layouts. The authors evaluated their approach on Malimg and MaleVis, reporting an accuracy of 96.10% and a Macro-F1 score of 96.00% on the MaleVis dataset. The study highlighted that optimizing custom lightweight modules is far more efficient than deploying heavy, pre-trained transfer networks. However, the architecture still struggled to accurately classify certain polymorphic malware families that use sophisticated code-packing techniques. A custom CNN leveraging alternative multi-scale attention mechanisms could be investigated to determine if it captures local obfuscated structures more effectively without adding heavy parameter depth.

2. Yao et al. (2025) proposed a Res2NeXt-based malware-classification framework incorporating a lightweight Adaptive Channel Attention (ACA) mechanism. The attention module emphasizes informative feature channels while reducing the influence of irrelevant information in malware RGB images. The authors evaluated the approach on MMCC and MaleVis, reporting accuracies of 99.26% and 98.04%, respectively. The study also used confusion matrices, t-SNE, and Grad-CAM visualizations to examine classification performance and learned image features. However, the specialized Res2NeXt architecture leaves room to investigate whether a simpler custom CNN with lightweight attention can achieve competitive results with fewer parameters and lower inference latency. A controlled comparison between the same CNN with and without attention would also help quantify the contribution of the attention mechanism.

3. Al-Anazi, Al-Madi, & Mahmoud (2025) presented a multi-stream deep transfer learning model that merges spatial feature extractions from VGG-16 and ResNet-50 architectures simultaneously. The model feeds raw visual inputs through parallel convolutional networks, using max-pooling and global average-pooling blocks to build a dense, concatenated 2048-dimensional joint feature representation. When evaluated across several benchmarks, the model achieved a notable accuracy of 98.01% on the MaleVis dataset. The study demonstrated the raw strength of multi-backbone ensembles in minimizing structural pattern losses. However, the model relies entirely on a brute-force architectural combination and lacks a dynamic attention mechanism to filter out irrelevant background byte code or non-informative padded regions. This leaves room to explore whether a single-stream custom CNN paired with an intentional spatial-channel attention layer can match this performance at a fraction of the computational and memory footprint.

4. Bavishi & Modi (2024) designed a dual-stage hybrid network, known as LeViT-MC, linking a fine-tuned DenseNet CNN backbone with a Vision Transformer encoder sequence to accelerate multi-class malware identification. The framework utilizes a specialized multi-head self-attention mechanism within the transformer blocks to compute global context and capture long-range byte-level structural dependencies across the transformed binary sequences. Evaluated directly on the 26-class MaleVis distribution, the framework achieved an outstanding multi-class classification accuracy of 96.60% while maintaining a highly efficient inference throughput. However, the dense interconnection of the Transformer heads and the hybrid backbone makes the network incredibly complex, presenting a significant barrier to explainable AI (XAI) mapping. This leaves an open gap to investigate whether a simpler, purely convolutional CNN with lightweight attention can maintain a competitive classification accuracy while offering clearer, more meaningful local feature interpretations through standard toolsets like Grad-CAM.

5. Nkrumah-Asante et al. (2024) outlined a Data-Efficient Image Transformer pipeline, named DeiT-MC, optimized specifically for robust malware classification without requiring massive pre-training datasets. The model utilizes multi-head self-attention layers to map global contextual dependencies, paired with a custom Hybrid GridBay Optimizer (HGBO) to carefully automate and stabilize hyperparameter tuning. The authors evaluated the model natively on the MaleVis dataset, reporting a classification accuracy of 94.00%. While the study proved that data-efficient

transformers can bypass traditional CNN inductive biases to mitigate code structural shifts, its performance falls noticeably below heavy convolutional alternatives. This performance difference reveals a distinct research gap: pure global self-attention heads often blur out localized, fine-grained byte-level visual textures. A controlled study into a custom convolutional CNN with targeted attention would help determine if localized kernels are fundamentally better suited for parsing pixel-level malware patterns than standard transformer blocks.

Summary Table

Paper	Year	Dataset / Protocol	Method	Attention Type	Main Result	Research Gap
[1] Al-Harbi & AlShorman	2026	MaleVis and Malimg	A-HRAB framework using standard convolutional networks)	Adaptive Hybrid Residual Attention (Channel + Spatial balancing	96.10% accuracy (96.00% Macro-F1) on MaleVis	Optimizing lightweight custom blocks struggles to correctly isolate structural code variants when sophisticated code-packing or multi-layer encryption tools are used by polymorphic variants.
[2] Yao et al.	2025	MaleVis and MMCC	Res2NeXt backbone architecture	Lightweight Adaptive Channel Attention (ACA)	98.04% accuracy on MaleVis (99.26% on MMCC)	The underlying Res2NeXt network relies heavily on high parameter density, which dramatically spikes inference latency and computational training overhead in constrained or local detection endpoints.
[3] Al-Anazi, Al-Madi, & Mahmoud	2025	MaleVis, Malimg, and Microsoft BIG 2015	Multi-stream feature fusion (Parallel VGG-16 and ResNet-50)	None (Traditional raw deep feature concatenation)	98.01% accuracy on MaleVis	Relying entirely on brute-force, dual-backbone architectural pooling lacks any localized attention to dynamically filter out dead binary padding

						code or noisy background visual artifacts.
[4] Bavishi & Modi	2024	MaleVis (26-class malware and benign distribution)	LeViT-MC hybrid framework (DenseNet CNN paired with a Vision Transformer)	Multi-Head Self-Attention within transformer sequence layers	96.60% multi-class accuracy on MaleVis	The structural density of the hybrid transformer heads introduces deep model layer abstraction, creating a severe obstacle for clean, interpretable local XAI heatmapping via standard Grad-CAM blocks.
[5] Nkrumah-Asante et al.	2024	MaleVis and MalNet-Image Tiny datasets	DeiT-MC pipeline driven by a Hybrid GridBay Optimizer (HGBO)	Multi-Head Self-Attention built natively into Vision Transformer	94.00% classification accuracy on MaleVis	Global self-attention transformer blocks inherently smooth over localized pixel features, causing the network to blur out highly granular, texture-based localized binary code structures.

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